

BIG DATA ANALYTICS & MACHINE LEARNING IN THE REAL WORLD

October 2025

Big Data Analytics

DATA STORYTELLING

Data Stories Gallery - Microsoft Fabric Community

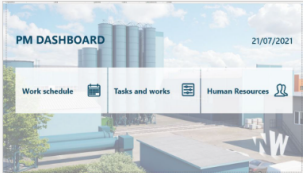
https://community.fabric.microsoft.com/t5/Data-Stories-Gallery/bd-p/DataStoriesGallery?sortBy=kudos

Microsoft | Fabric Community Forums Inspiration Ideas Communities Blogs Learning Support

Data Stories Gallery

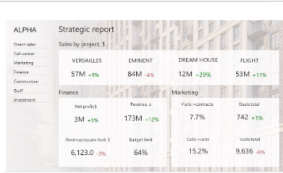
Recently Posted Featured **Top Kudos** All Data Stories

[Submit your data story](#) [Options](#)



Construction management

a9126030767 557



Property developer's sales analytics

alexkolokolov 543



Airport Authority Performance Summary

mmatey 335



Trade in Europe

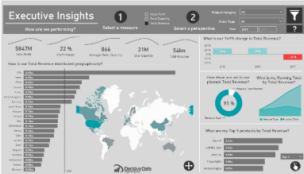
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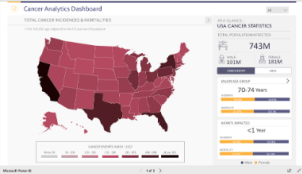
Global Super Store



Climate Change is Real



Executive Insights by Decisive Data



Cancer Analytics

MEASUREMENT FRAMEWORK

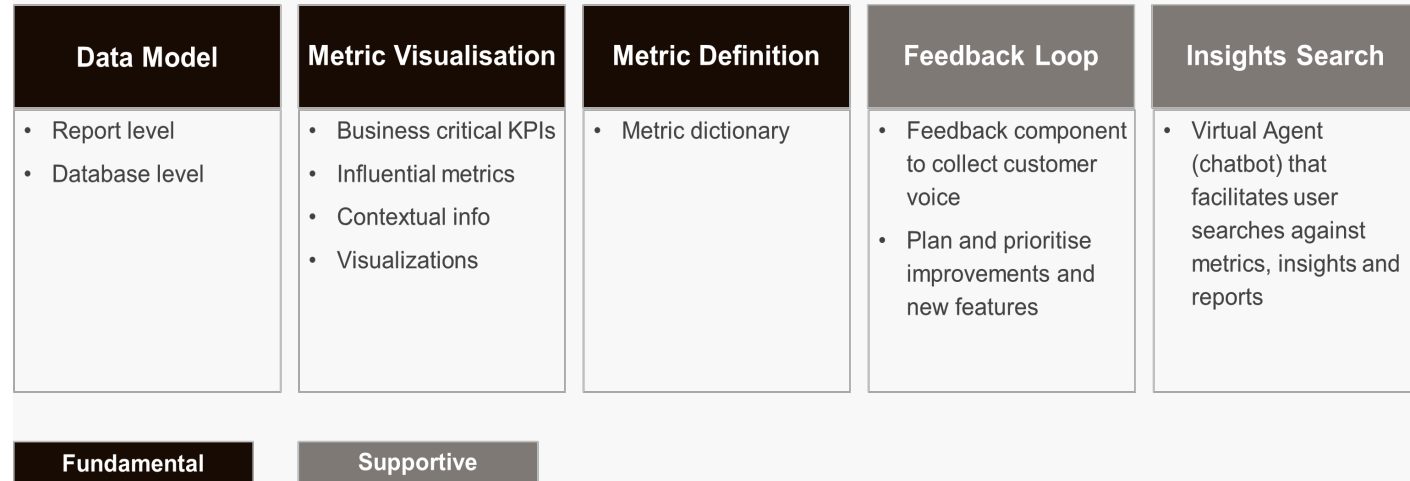
Tier 1: Business Outcomes	Audience: CMO or CDO Purpose: Assess Digital Commerce Results and Contribution to Business Goals.
Tier 2: Strategic Metrics	Audience: VP of Marketing and/or VP of Digital Commerce Purpose: Identify and Diagnose Issues Affecting Digital Commerce Results.
Tier 3: Operational Metrics	Audience: Category, Account, Platform or Program Director Purpose: Manage Marketing and Merchandising Based on Customer Behavior.
Tier 4: Tactical Metrics	Audience: Marketing Managers Purpose: Optimize Marketing Effectiveness for Impact on Digital Commerce Results.

AN EXAMPLE MEASUREMENT FRAMEWORK FOR E-COMMERCE

	Strategic Aim	Improving E-commerce growth and ROI			
	Key Result Area	Demand Generation	Digital Sales	Customer Experience	Brand & Reputation
	Strategic Metric	Digital Commerce Revenue Growth	E-commerce Revenue	Customer Lifetime Value	Loyalty CAST (Customer Satisfaction Score)
		Cross-sell / Upsell Ratio	E-commerce Revenue Profit Margin	Customer Acquisition Cost (CAC)	
		Sales Conversion Rate	E-commerce ReSales as a % of Total Sales		
Operational Metric	Operational Metric	Conversion Rate by Product Category	No. of Sales Transactions	Customer Acquisition Rate	Sentiment Rating
		Purchase Frequency	Avg Qty per Transaction	Retention / Renewal Rate	Rating by Product Category
		Repeat Visitor Rate	Avg Order Value (AOV)	CAC by Product Category	User Stickiness
		Visit to Page to Cart Conversion	Shopping Cart Abandonment Rate		
Tactical Metric	Tactical Metric	Click Through Rate (CTR)	Avg Time on Site	Registration Rate	Monthly Repeat Users
		Bounce Rate	Conversation by Channel	Engagement Rate by Product Category	Daily Avg Users
		Traffic by Channel	AOV by Marketing Campaign	Scroll Depth	
		Pageviews per Visit			
		Avg Dwell Time (login-to-checkout) by Product			
		Traffic by Segment			
		Traffic by Device			

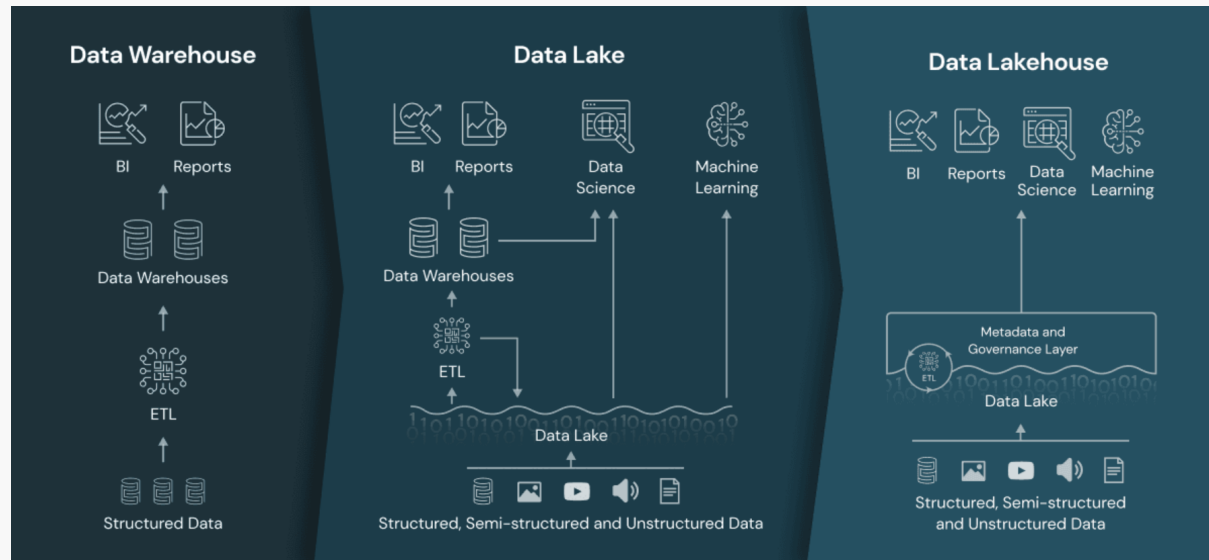
ANALYTICS SOLUTION FRAMEWORK

- Building blocks of a general analytics solution framework
- Visualisation - key to present metrics and KPIs
- Machine learning for creating data models

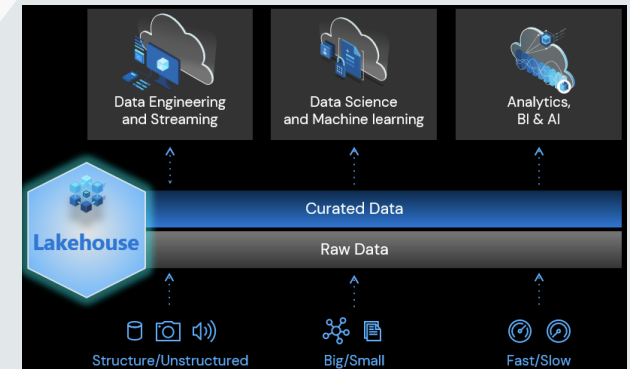


DATA LAKEHOUSE

- Lakehouse brings the best of data lake and data warehouse in a single unified data platform
- A single source of truth for data of all types and formats
- Integrated capabilities for storage, ETL/ELT, processing in parallel for scale, analytics, and machine learning



- There is no separate data warehouse and data lake. The data lake is the data warehouse.
- Delta Lake – is the secret sauce of the Lakehouse pattern and looks to resolve the limitations of data lake.



GARTNER'S MAGIC QUADRANT FOR ANALYTICS AND INTEGRATION PLATFORMS

Magic Quadrant for Analytics & Business Intelligence Platforms



Magic Quadrant for Data Integration Tools

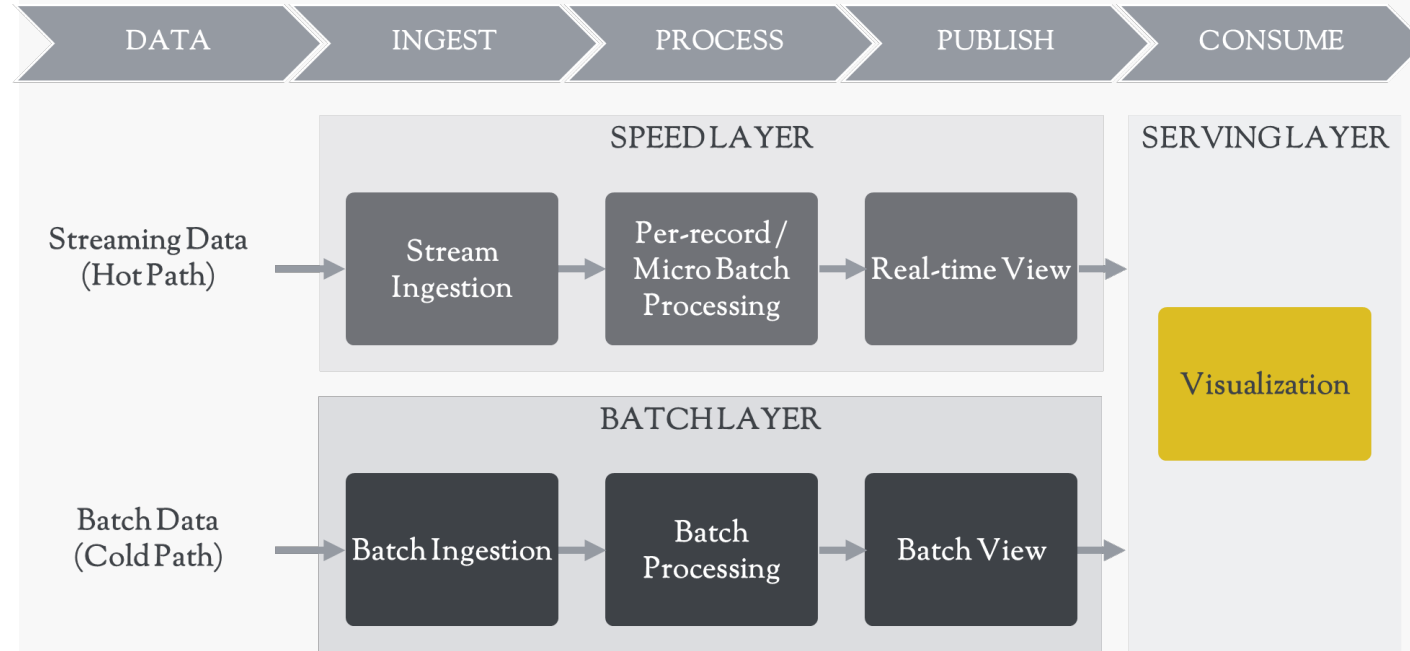


EVALUATION OF POPULAR CLOUD-BASED ANALYTICS PLATFORMS

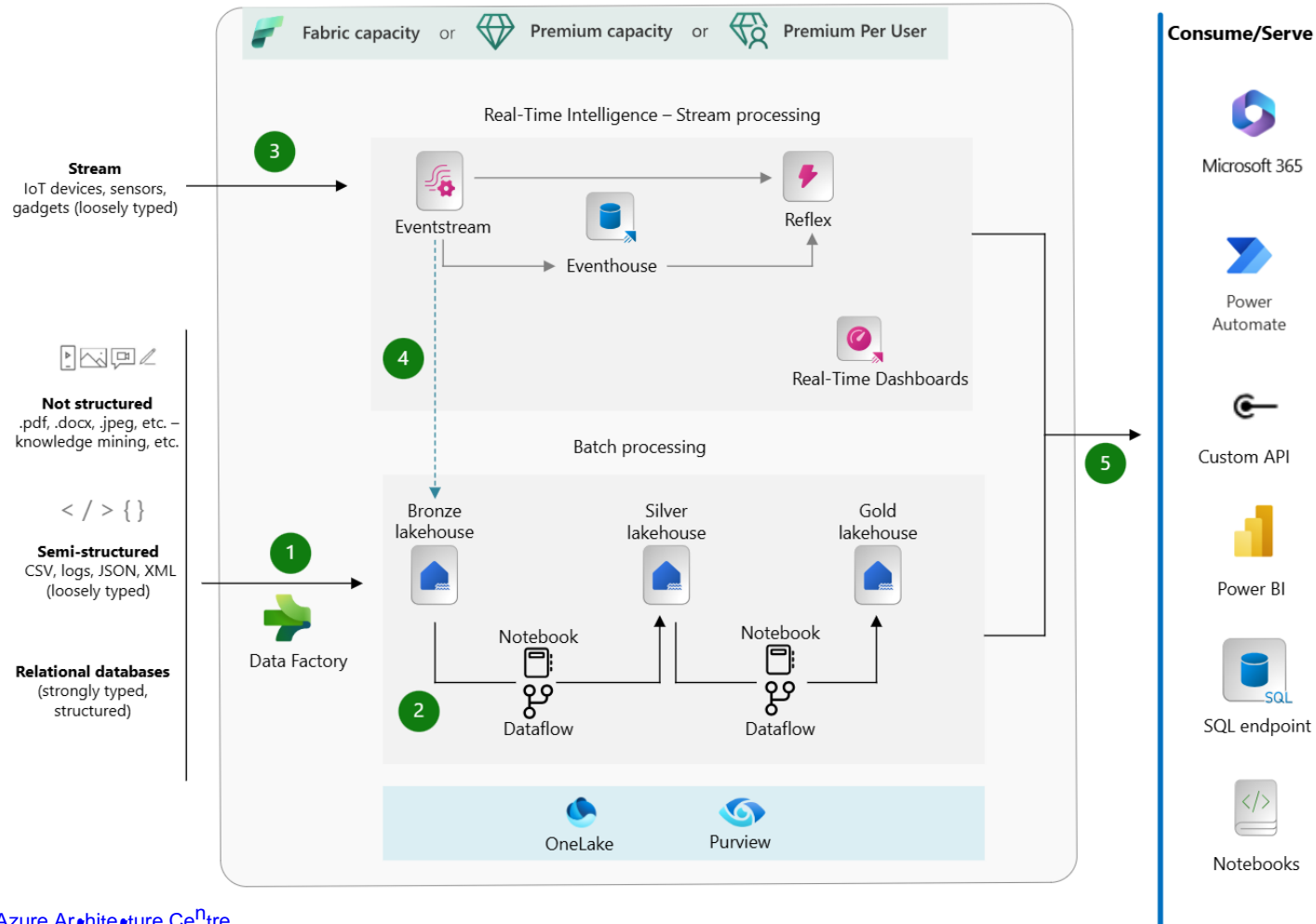
	Pillar	Microsoft Fabric	AWS Native Stack	Databricks Lakehouse	Snowflake
Excellent	Platform cohesion	Single SaaS surface (OneLake) with unified experiences across ingestion, engineering, warehousing, real-time, and Power BI. Minimal stitching.	Best-of-breed components but user-assembled; many services to integrate and operate.	Cohesive for lakehouse workloads; still pairs with external BI/ETL/governance in many shops.	Excellent warehouse core; relies on partner tools for many upstream/downstream steps.
	Governance & security	Central governance, OneLake items, fine-grained security; tight integration with Microsoft Purview and M365 labels.	Powerful (IAM, Lake Formation) but policy design and cross-service enforcement are on you.	Unity Catalog gives consistent authentication/authorisation, lineage, and governance across workspaces.	Mature warehouse governance (RBAC, masking, tags); outside-warehouse governance handled via integrations.
Strong	Dev lifecycle & deployment automation	Built-in deployment pipelines, environment promotion, semantic model versioning; opinionated yet simple.	Robust CI/CD options but require nontrivial assembly and standards.	Workflows, repos, Delta Live Tables; strong for data engineering, some assembly for BI release.	Tasks/Streams and external CI/CD work; less opinionated for multi-env analytics programs.
Moderate	BI & semantic layer integration	Native Power BI + centralized semantic models/metrics; end-to-end from data to decisions.	QuickSight is improving; most enterprises use 3 rd party BI — less seamless semantics.	Connects well to Power BI/Tableau; no native, first-class BI semantic layer equivalent.	Strong with external BI; emerging native metrics but not the same as embedded Power BI.
Limited	Streaming & real-time	Real-Time analytics (KQL/RTI) built in; event ingestion integrated with the platform.	Kinesis family is best-in-class breadth; MSK/EMR also options. Requires stitching.	Structured Streaming and Auto Loader are excellent for continuous pipelines.	Snowpipe Streaming and Tasks are solid for near-real-time; fewer native stream operators.
	AI/ML readiness	Notebooks, Python/SQL, integration with Azure ML, and Copilot experiences in product.	SageMaker + deep AI service catalog; top-tier breadth and depth.	First-class ML/GenAI workflows; strong open-source DNA (Spark/MLflow).	Snowpark (Python/Java/Scala) and in-db ML; strong for SQL/feature engineering, less open than DBX.
	Cost model & operational overhead	SaaS capacity model; low infra burden, fast time-to-value, predictable admin surface.	Fine-grained cost control, but higher ops tax; engineering time to integrate/operate many services.	Powerful but can require expert tuning and cluster strategy; ops burden > SaaS.	Serverless consumption with simple ops; cost can sprawl without governance/warehousing best practices.

BIG DATA ARCHITECTURE

Lambda Architecture -
a classical approach to
big data processing



FABRIC LAKEHOUSE ARCHITECTURE



TYPICAL USE CASES OF FABRIC LAKEHOUSE

Predictive Modelling & Forecasting

- Sales, demand, and revenue forecasting
- Predictive maintenance for IoT or manufacturing equipment
- Customer churn prediction and retention optimization
- Workforce or capacity planning

Anomaly & Fraud Detection

- Real-time fraud detection in transactions or payments
- Sensor and equipment fault detection
- Log or system performance anomaly monitoring
- Outlier spotting in financial or operational data

Customer & Product Segmentation

- Behavioural segmentation for marketing & personalization
- Product recommendation systems
- Cohort and lifecycle analysis
- Market basket or cross-sell analysis

Natural Language & AI-Powered Insights

- Sentiment analysis and text classification
- Document or image intelligence (tagging, summarization)
- Generative AI for automated reports or insights
- Knowledge mining from PDFs, docs, and emails

Optimization & Simulation

- Supply chain and logistics route optimization
- Resource and workforce scheduling
- Financial portfolio or pricing optimization
- Scenario modelling and what-if simulations

Real-Time Decisioning & Automation

- Dynamic pricing or offer optimization in real-time
- Instant loan or credit approval based on live data
- Operational triggers via Reflex or Power Automate
- Continuous feedback loops for adaptive systems

Operationalized Machine Learning (MLOps)

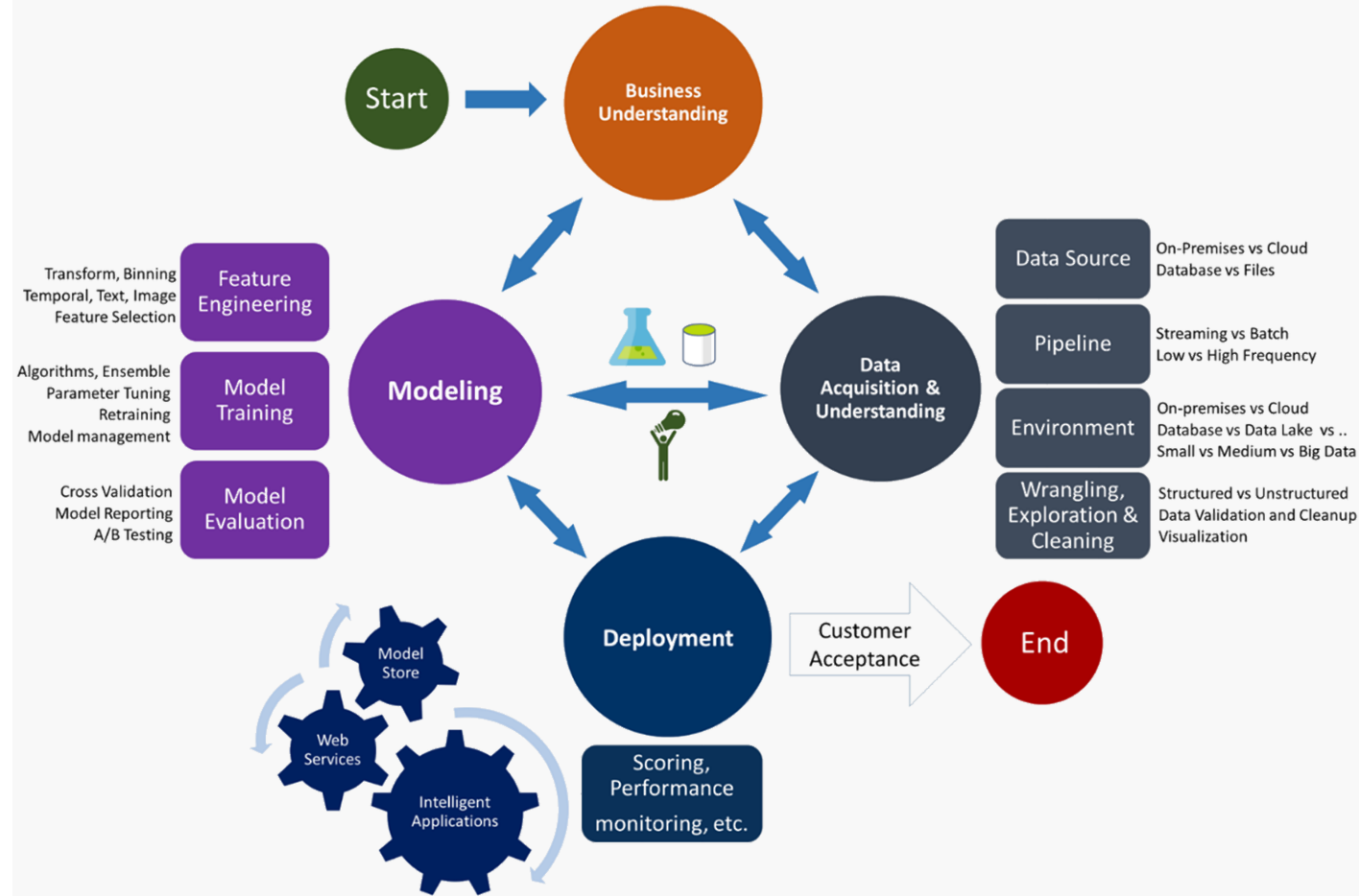
- Automated model training and retraining pipelines
- Model deployment and monitoring directly in Fabric
- Real-time scoring with Eventstream or APIs
- Version control and performance tracking of models

Data-Driven Innovation & Experimentation

- A/B testing and uplift modelling for marketing experiments
- R&D analytics for product development
- AI-assisted hypothesis testing on unified datasets
- Discovery of new correlations or trends in Gold Lakehouse

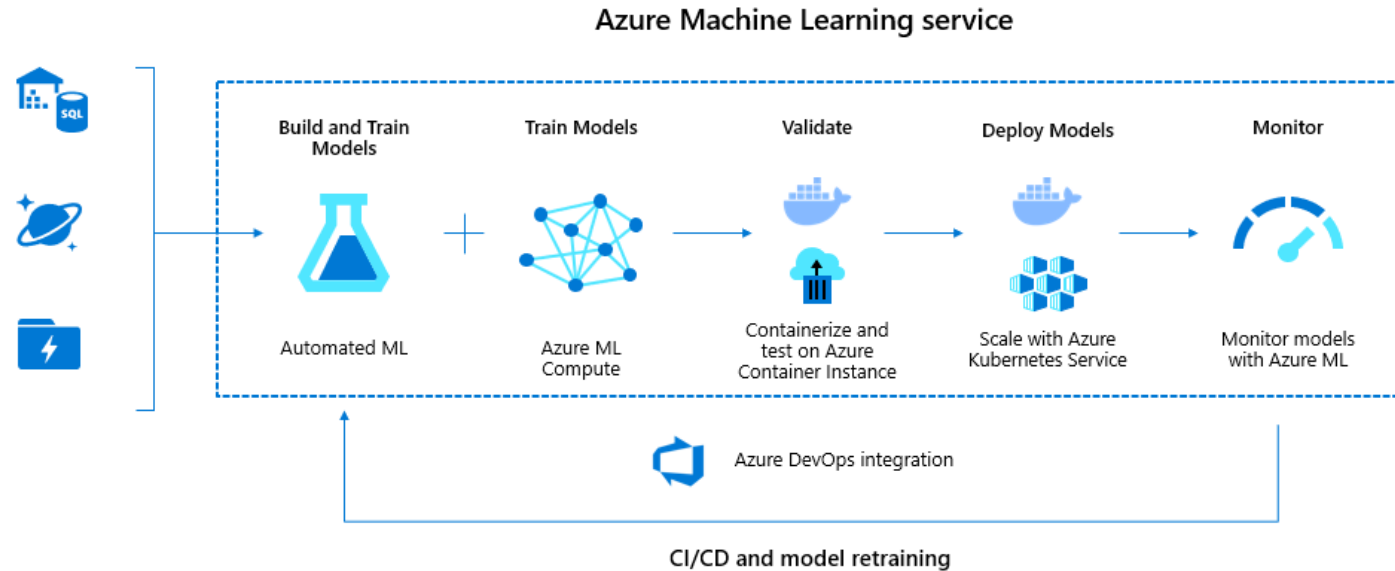
Machine Learning

MACHINE LEARNING LIFECYCLE

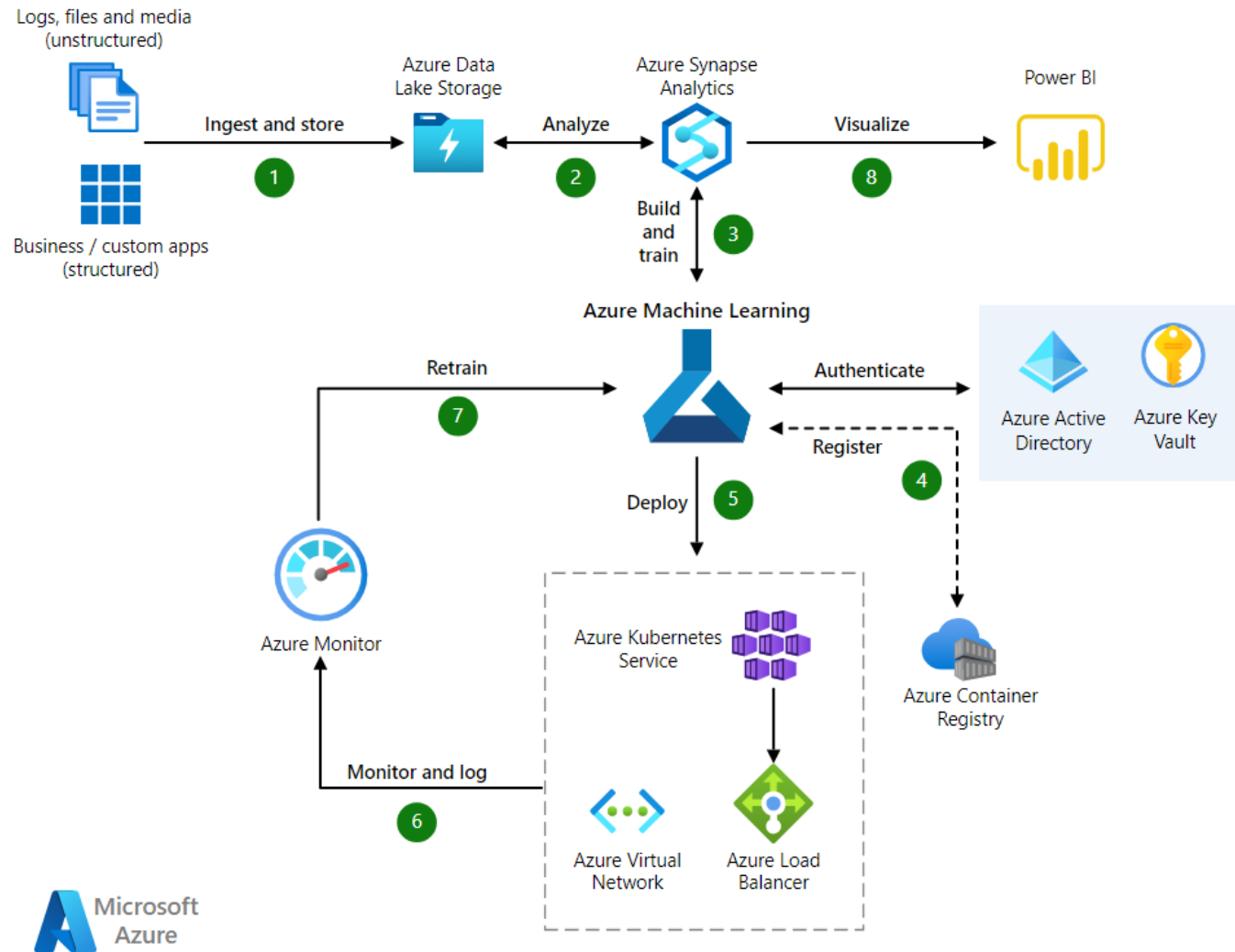


AZURE MACHINE LEARNING LIFECYCLE

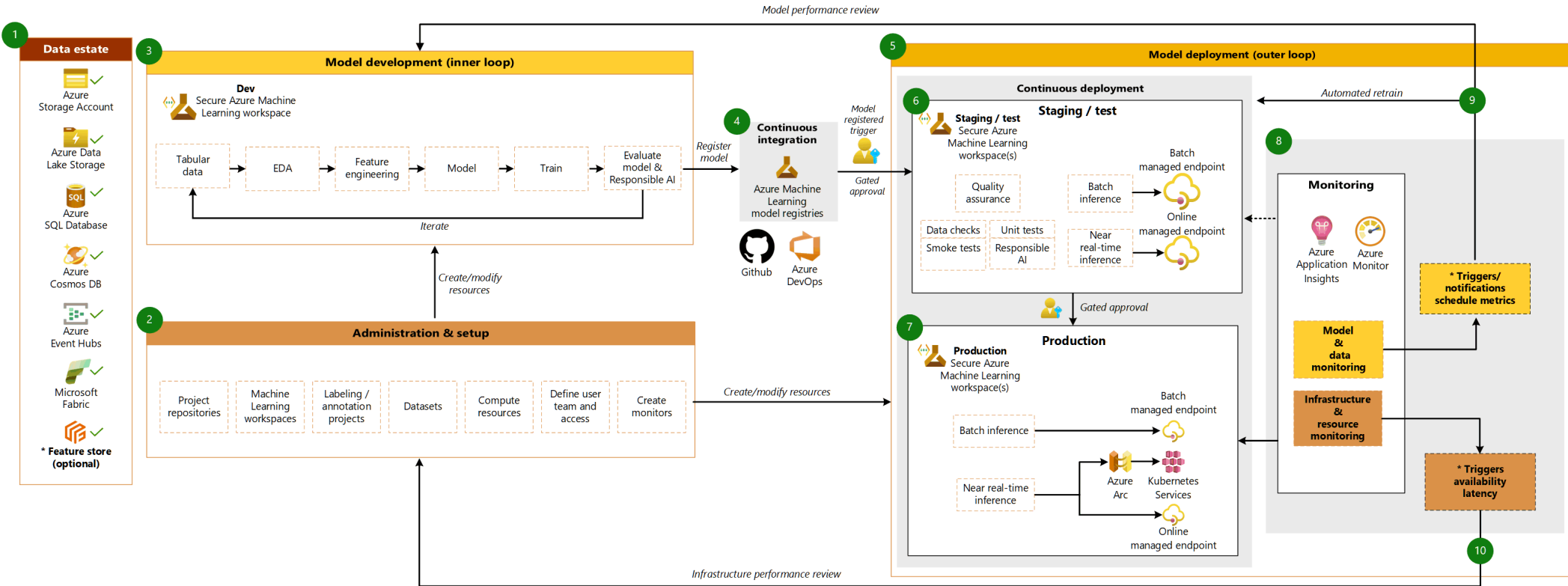
Implementation with
Azure data services



Azure Machine Learning Architecture



AZURE MACHINE LEARNING WORKFLOWS



Legend: Data engineer Data scientist Machine learning engineer Infrastructure team * Future accelerator implementation ✓ Recommended practice

MACHINE LEARNING IN HEALTHCARE



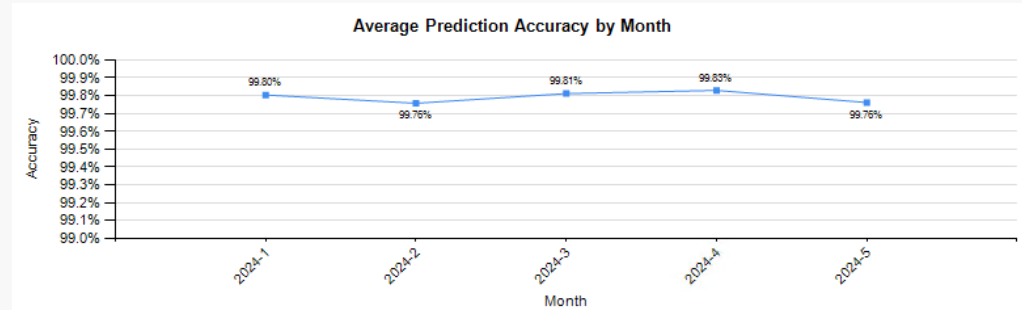
Source: [ML iⁿ Healthcare](#)

MACHINE LEARNING IN EMERGENCY HEALTHCARE

Use machine learning
to predict what type of
ambulance should be
sent to an incident

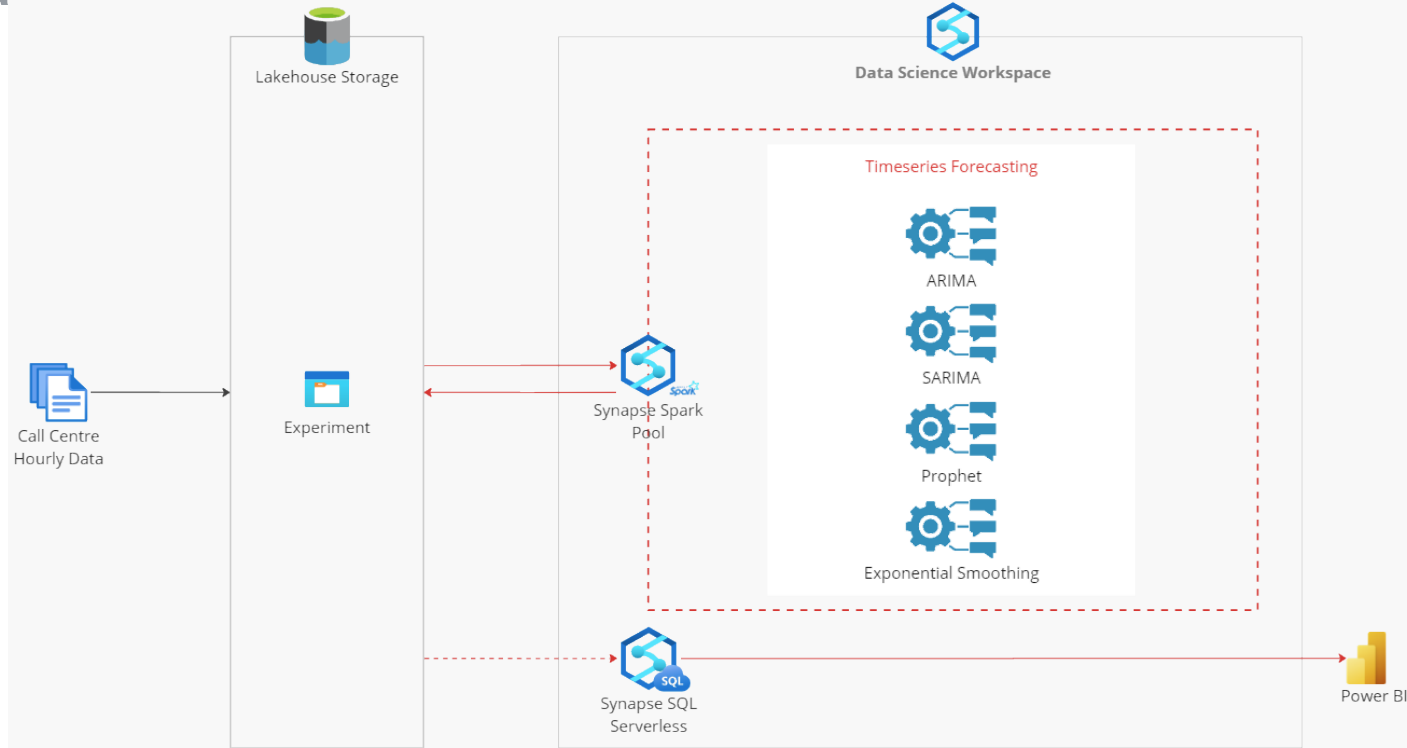


- Ensemble learning
- Catboost, decision tree
- 42 features
- Prediction accuracy: 99.2%
- Ongoing model maintenance plan



CALL CENTRE PERFORMANCE & WORKFORCE FORECASTING

- Leveraged time series forecasting algorithms to predict key call centre performance metrics, including inbound call volume, speed-to-answer, handling time, agent utilisation, and lost time
- The forecasts informed workforce planning by estimating the number of call handlers required per shift to meet service-level targets.



- Used NLP-driven machine learning to detect and prioritize incidents causing death, permanent harm, or severe temporary harm.
- The solution summarizes each incident report into key representative keywords, predicts severity scores, and classifies incidents based on the highest keyword severity identified.

